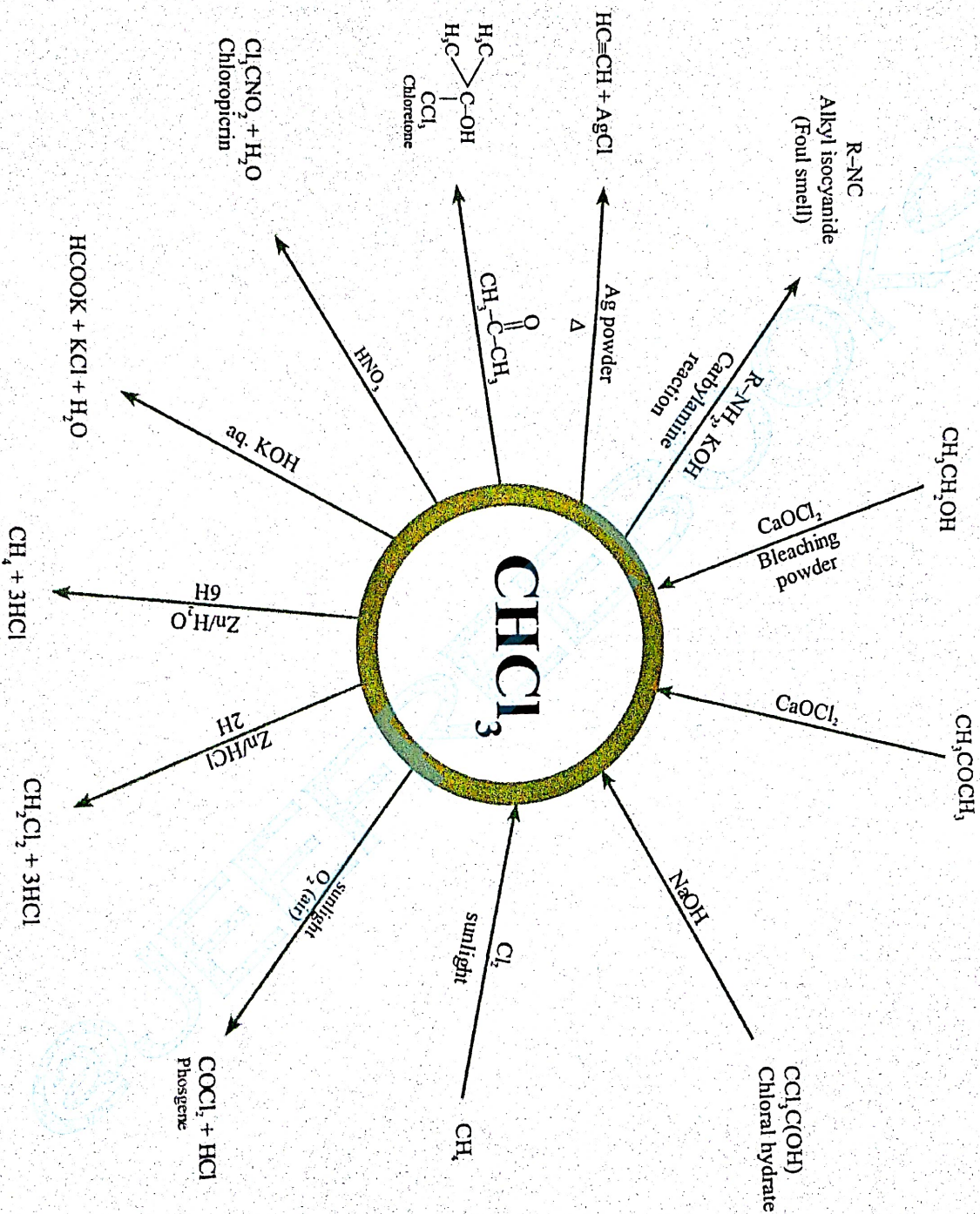
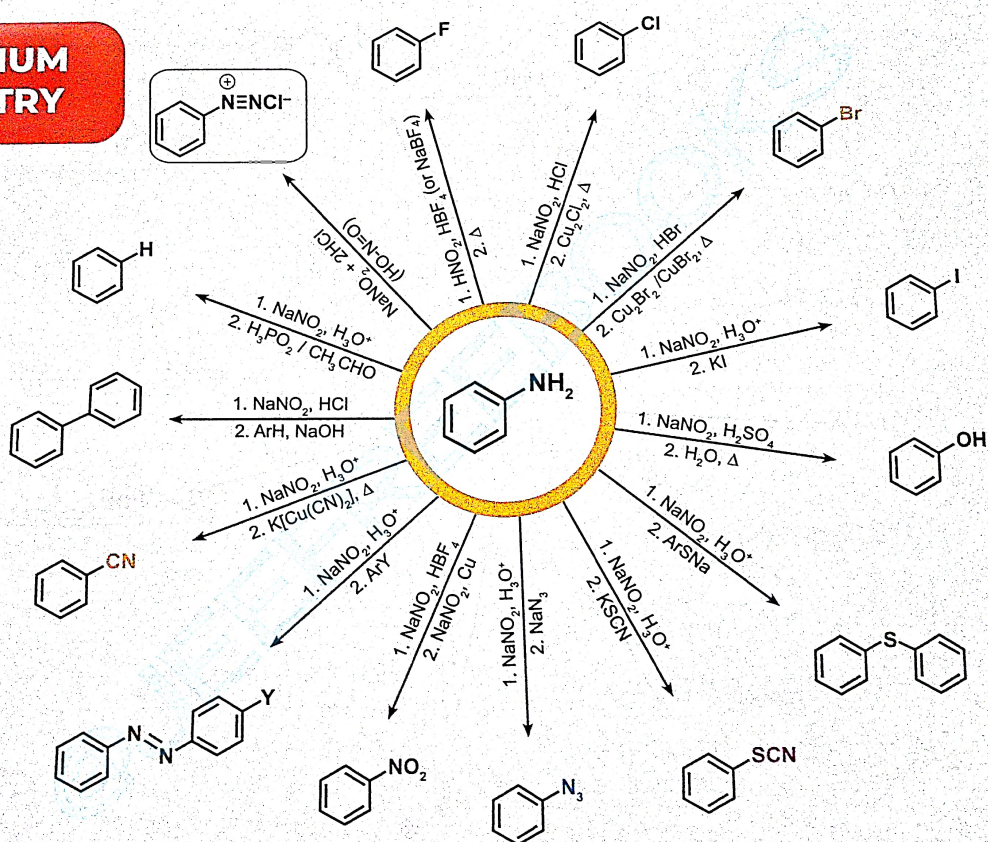


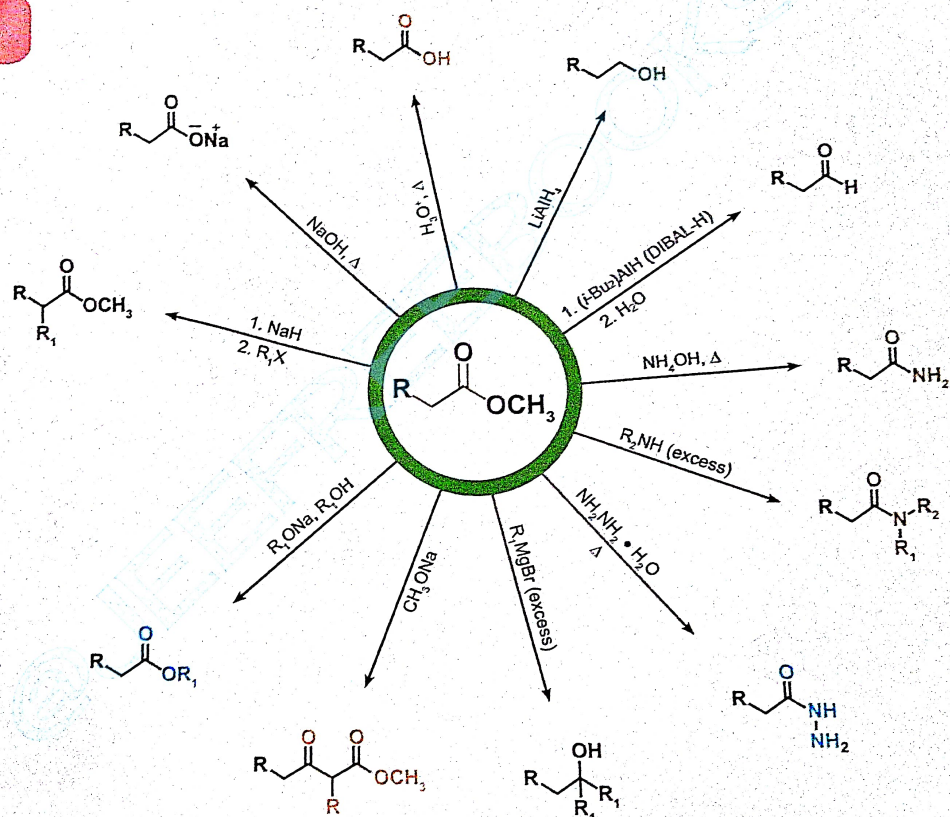
CHLOROFORM



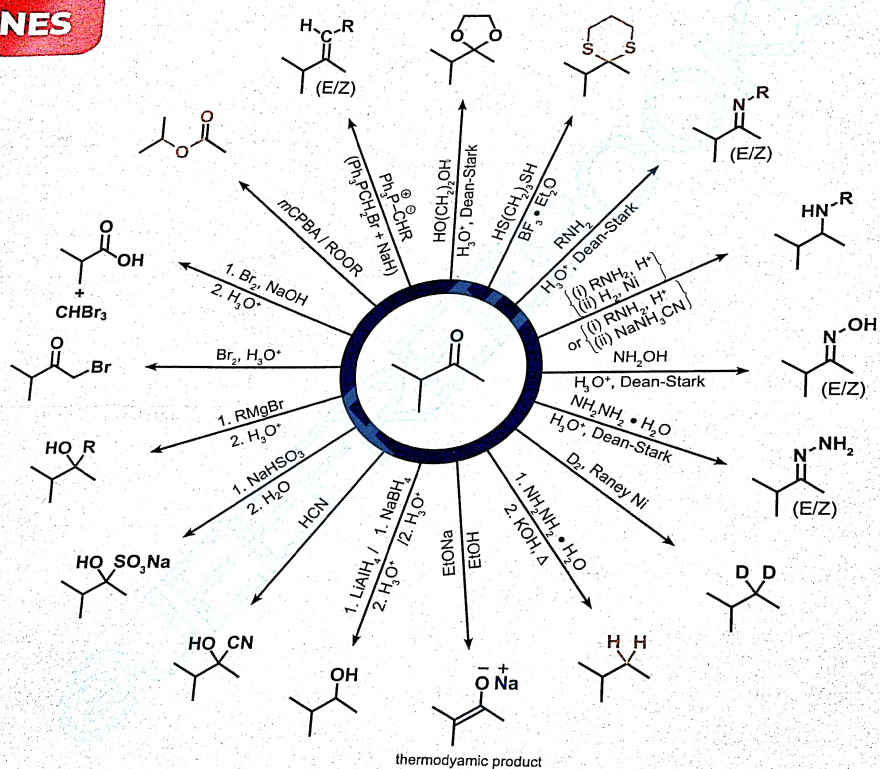
53



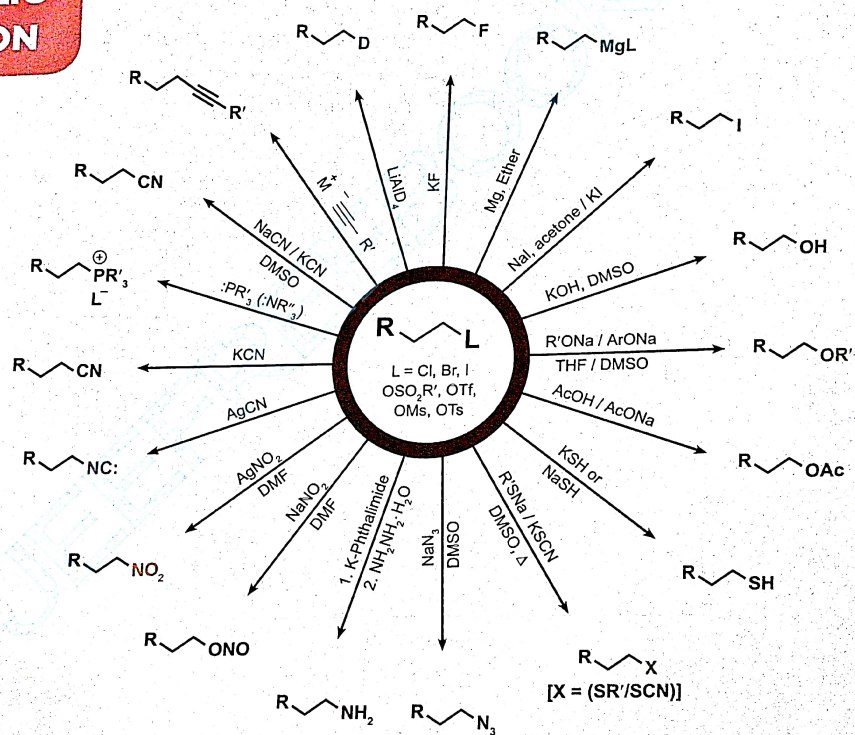
ESTERS



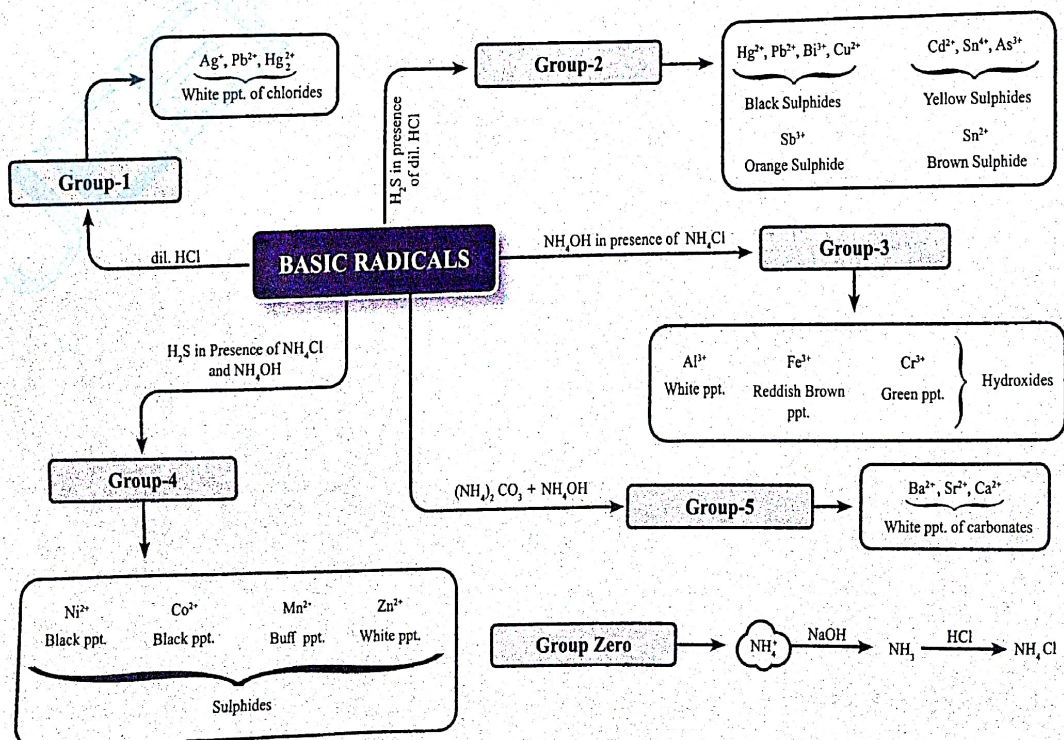
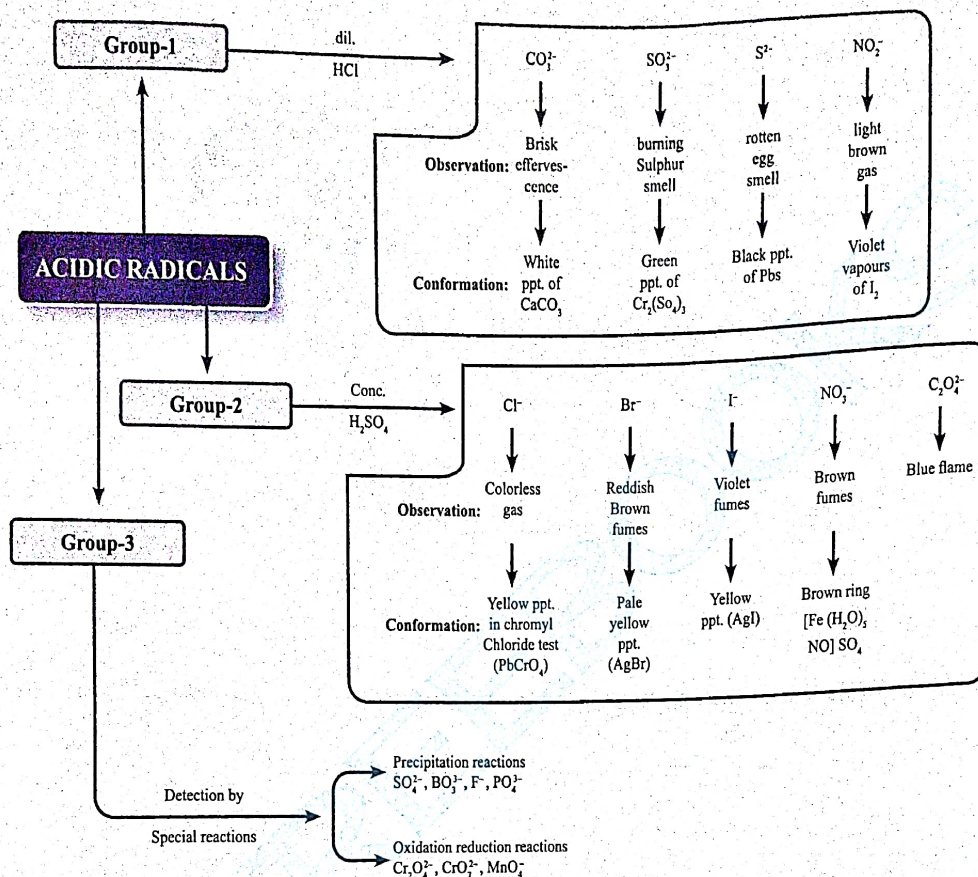
KETONES



NUCLEOPHILIC SUBSTITUTION



SALT ANALYSIS



MIND MAP

MATHEMATICS



RELATIONS



"Mathematics is, in its way, the poetry of logical ideas."

(1) Types of Relations

1. Empty Relation

A relation in which no element of A is related to any other element of A, i.e., $R = \emptyset \subset A \times A$.

2. Universal Relation

A relation in which each element of A is related to every element of A, i.e., $R = A \times A$.

3. Identity Relation

A relation in which each element is related to itself only. $I = \{(a, a), a \in A\}$

4. Reflexive Relation:

$(a, a) \in R$, for every $a \in A$.

5. Symmetric Relation:

$(a, a_1) \in R$ implies that $(a_1, a) \in R$ for all $a, a_1 \in A$.

6. Transitive Relation:

$(a, a_1) \in R$ & $(a_1, a_2) \in R$ implies that $(a, a_2) \in R$, for all $a, a_1, a_2 \in A$.

7. Equivalence Relation:

A relation R in a set A is said to be an equivalence relation if R is reflexive, symmetric & transitive.

8. Inverse Relation

Inverse relation of R from A to B, denoted by R^{-1} , is a relation from B to A is defined by $R^{-1} = \{(b, a) : (a, b) \in R\}$.

9. Asymmetric Relation

$(x, y) \in R \Rightarrow (y, x) \notin R$

10. Antisymmetric: A relation is antisymmetric if:

• For all $x, y \in X$ $\{ (x, y) \in R \& (y, x) \in R \} \Rightarrow x = y$
• For all $x, y \in X$ $\{ (x, y) \in R \& x \neq y \} \Rightarrow (y, x) \notin R$

11. Irreflexive

R is irreflexive iff $\forall a \in A, (a, a) \notin R$

12. Partial order relation

R is a partial order, if R is Reflexive, Antisymmetric and Transitive.

2. EXAMPLE:

$A = \{1, 2, 3, 4\}$. Identify the properties of relations.

$R_1 = \{(1, 1), (2, 2), (3, 3), (2, 1), (4, 3), (4, 1), (3, 2)\}$

$R_2 = A \times A$, $R_3 = \emptyset$, $R_4 = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$

$R_5 = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (4, 3), (3, 4)\}$

Relation	Reflexive	Symmetric	Asymmetric	Antisymmetric	Irreflexive	Transitive
R_1	X	X	X	✓	X	X
R_2	✓	✓	X	X	X	✓
R_3	X	✓	✓	✓	✓	✓
R_4	✓	✓	X	✓	X	✓
R_5	✓	✓	X	X	X	✓

NOTE

If $A = \{1, 2\}$, a relation $R = \{(1, 2)\}$ on A is a transitive relation. using the similar argument a relation $R = \{(x, y) : x \text{ is wife of } y\}$ is transitive, where as $R = \{(x, y) : x \text{ is father of } y\}$ is not transitive.

3. PROPERTIES

1.

R is not reflexive does not imply R is irreflexive. Counter example:

$A = \{1, 2, 3\}$, $R = \{(1, 1)\}$

2.

R is asymmetric implies that R is irreflexive. By definition, for all $a, b \in A$, $(a, b) \in R$ and $(b, a) \notin R$. This implies that for all $(a, b) \in R$, $a \neq b$. Thus, for all $a \in A$, $(a, a) \notin R$. Therefore, R is irreflexive.

3.

R is not symmetric does not imply R is antisymmetric. Counter example:

$A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 3), (3, 2)\}$

4.

R is not symmetric does not imply R is asymmetric. Counter example:

$A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 2)\}$

5.

R is not antisymmetric does not imply R is symmetric. Counter example:

$A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 3), (3, 2)\}$

6.

R is reflexive implies that R is not asymmetric. By definition, for all $a \in A$, $(a, a) \in R$.

This implies that, both (a, b) and (b, a) are in R when $a = b$. Thus, R is not asymmetric.

(4) COUNTING OF RELATION

Number of relations from set A to B = 2^{mn} , where $|A| = m, |B| = n$

Number of Identity relation on a set with 'n' elements = 1

Number of reflexive relation set on a set with 'n' elements = $2^{n(n-1)}$

Number of Symmetric relation set on a set with 'n' elements = $2^{n(n+1)/2}$

The number of antisymmetric binary relations possible on A is $2^n \cdot 3^{(n^2-n)/2}$

The number of binary relation on A which are both symmetric and antisymmetric is 2^n .

The number of binary relation on A which are both symmetric and asymmetric is 1.

The number of binary relation which are both reflexive and antisymmetric on the set A is $3^{(n^2-n)/2}$

The number of asymmetric binary relation possible on the set A is $3^{(n^2-n)/2}$

There are at least 2^n transitive relations (lower bound) and at most $2^{n^2 - 2 \frac{n^2-n}{2} + 1}$ (upper bound)

5. OPERATION ON RELATIONS:

$$1. R_1 - R_2 = \{(a, b) \mid (a, b) \in R_1 \text{ and } (a, b) \notin R_2\}$$

$$2. R_2 - R_1 = \{(a, b) \mid (a, b) \in R_2 \text{ and } (a, b) \notin R_1\}$$

$$3. R_1 \cup R_2 = \{(a, b) \mid (a, b) \in R_1 \text{ or } (a, b) \in R_2\}$$

$$4. R_1 \cap R_2 = \{(a, b) \mid (a, b) \in R_1 \text{ and } (a, b) \in R_2\}$$

PROPERTIES

1) If R_1 and R_2 are reflexive, and symmetric, then $R_1 \cup R_2$ is reflexive, and symmetric.

2) If R_1 is transitive and R_2 is transitive, then $R_1 \cup R_2$ need not be transitive.

counter example: Let $A = \{1, 2\}$ such that $R_1 = \{(1, 2)\}$ and $R_2 = \{(2, 1)\}$. $R_1 \cup R_2 = \{(1, 2), (2, 1)\}$ and $(1, 1) \notin R_1 \cup R_2$ implies that

$R_1 \cup R_2$ is not transitive.

3) If R_1 and R_2 are equivalence relations, then $R_1 \cap R_2$ is an equivalence relation.

4) If R_1 and R_2 are equivalence relations on A ,

- $R_1 - R_2$ is not an equivalence relation (reflexivity fails).
- $R_1 - R_2$ is not a partial order (since $R_1 - R_2$ is not reflexive).
- $R_1 \oplus R_2 = R_1 \cup R_2 - (R_1 \cap R_2)$ is neither equivalence relation nor partial order (reflexivity fails)

5) The union of two equivalence relation on a set is not necessarily an equivalence relation on the set.

6) The inverse of a equivalence relation R is an equivalence relation.

6. COMPOSITON OF RELATIONS

Let $R_1 \subseteq A \times B$ and $R_2 \subseteq B \times C$, Composition of R_2 on R_1 , denoted as $R_1 \circ R_2$ or simply $R_1 R_2$ is

$$R_1 R_2 = \{(a, c) \mid a \in A, c \in C \wedge \exists b \in B \text{ such that } (a, b) \in R_1, (b, c) \in R_2\}$$

NOTE

$$R_1 (R_2 \cap R_3) \subseteq R_1 R_2 \cap R_1 R_3$$

$$R_1 (R_2 \cup R_3) = R_1 R_2 \cup R_1 R_3$$

$$R_1 \subseteq A \times B, R_2 \subseteq B \times C, R_3 \subseteq C \times D, (R_1 R_2) R_3 = R_1 (R_2 R_3)$$

$$(R_1 \circ R_2)^{-1} = R_2^{-1} \circ R_1^{-1}$$

7. EQUIVALENCE CLASS

Equivalence class of $a \in A$ is defined as $[a] = \{x \mid (x, a) \in R\}$, that is all the elements related to a under the relation R .

Example

E =Even integers, O =odd integers.

- (i) All elements of E are related to each other and all elements of O are related to each other.
- (ii) No element of E is related to any element of O and vice-versa.
- (iii) E and O are disjoint and $Z = E \cup O$

The subset E is called the equivalence class containing zero and is denoted by $[0]$.

Properties: consider an equivalence relation R defiend on a set A .

$$1. \bigcup_{a \in A} [a] = A$$

$$2. \text{For every } a, b \in A \text{ such that } a \in [b], a \neq b \text{ it follows that } [a] = [b]$$

$$3. \sum_{x \in A} |[x]| = |R|$$

$$4. \text{For any two equivalence class } [a] \text{ and } [b], \text{ either } [a] = [b] \text{ or } [a] \cap [b] = \emptyset$$

$$5. \text{For all } a, b \in A, \text{ if } a \in [b] \text{ then } b \in [a]$$

$$6. \text{For all } a, b, c \in A, \text{ if } a \in [b] \text{ and } b \in [c], \text{ then } a \in [c]$$

$$7. \text{For all } a \in A, [a] \neq \emptyset$$

Congruence modulo n given by $a \equiv b \pmod{n}$ if and only if n divides $(a - b)$.

"IN MATHEMATICS
THE ART OF PROPOSING A QUESTION
MUST BE HELD OF HIGHER VALUE THAN SOLVING IT."
- Georg Cantor

FUNCTION

1 Classification of function

01. Constant function

$f(x) = k$, k is a constant.

02. Identity function

The function $y = f(x) = x$, $\forall x \in \mathbb{R}$
Here domain & Range both $\mathbb{R}$

03. Polynomial function

$y = f(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_n$, n is non negative integer, a_i are real constants. Given $a_0 \neq 0$, n is the degree of polynomial function

There are two polynomial functions, $f(x) = 1 + x^n$ & $f(x) = 1 - x^n$ satisfying the relation: $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ where 'n' is a positive Integer.

04. Rational function

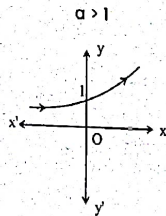
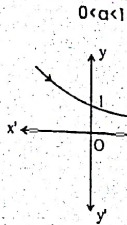
It is defined as the ratio of two polynomials.

$$f(x) = \frac{P(x)}{Q(x)} \text{ provided } Q(x) \neq 0$$

Dom $\{f(x)\}$ is all real numbers except when denominator is zero [i.e., $Q(x) \neq 0$]

2 Exponential function

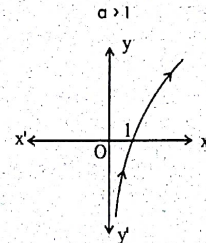
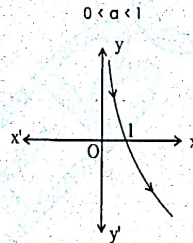
$$f(x) = a^x, a > 0, a \neq 1.$$



Domain = $\mathbb{R}$, Range = $(0, \infty)$

3 Logarithmic function

$$f(x) = \log_a x [a > 0, a \neq 1]$$



Domain = $(0, \infty)$, Range = $\mathbb{R}$

Properties of Log. Functions

1. $\log_a(xy) = \log_a |x| + \log_a |y|$, where $a > 0, a \neq 1$ and $xy > 0$

2. $\log_a x = \frac{1}{\log_x a}$ for $a > 0, a \neq 1$ and $x > 0, x \neq 1$

3. $\log_a \left(\frac{x}{y}\right) = \log_a |x| - \log_a |y|$, where $a > 0, a \neq 1$ and $\frac{x}{y} > 0$

4. $\log_a (x^n) = n \log_a |x|$, where $a > 0, a \neq 1$ and $x^n > 0$

5. $\log_a x^m = \frac{m}{n} \log_a |x|$, where $a > 0, a \neq 1$ and $x > 0$

6. $x^{\log_a y} = y^{\log_a x}$ where $x > 0, y > 0, a > 0, a \neq 1$

7. If $a > 1$, then the values of $f(x) = \log_a x$ increase with the increase in x .
i.e. $x < y \Leftrightarrow \log_a x < \log_a y$

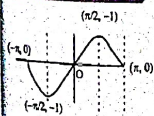
$$\text{Also, } \log_a x \begin{cases} < 0 & \text{for } 0 < x < 1 \\ = 0 & \text{for } x = 1 \\ > 0 & \text{for } x > 1. \end{cases}$$

8. If $0 < a < 1$, then the values of $f(x) = \log_a x$ decrease with the increase in x .
i.e. $x < y \Leftrightarrow \log_a x > \log_a y$

$$\text{Also, } \log_a x \begin{cases} > 0 & \text{for } 0 < x < 1 \\ = 0 & \text{for } x = 1 \\ < 0 & \text{for } x > 1 \end{cases}$$

4 Trigonometric Functions

Sine function

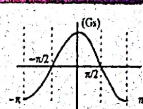


$$f(x) = \sin x$$

$$\text{Dom } (f) = \mathbb{R}$$

$$\text{Ran } (f) = [-1, 1]$$

Cosine function

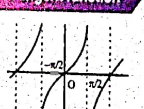


$$f(x) = \cos x$$

$$\text{Dom } (f) = \mathbb{R}$$

$$\text{Ran } (f) = [-1, 1]$$

Tangent function

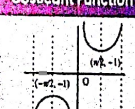


$$f(x) = \tan x$$

$$\text{Dom } (f) = \mathbb{R} - \left\{ \frac{(2n+1)\pi}{2}, n \in \mathbb{Z} \right\}$$

$$\text{Ran } (f) = \mathbb{R}$$

Cosecant function

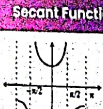


$$f(x) = \text{cosec } x$$

$$\text{Dom } (f) = \mathbb{R} - \{n\pi, n \in \mathbb{Z}\}$$

$$\text{Ran } (f) = \mathbb{R} - (-1, 1)$$

Secant function

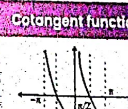


$$f(x) = \sec x$$

$$\text{Dom } (f) = \mathbb{R} - \left\{ \frac{(2n+1)\pi}{2}, n \in \mathbb{Z} \right\}$$

$$\text{Ran } (f) = \mathbb{R} - (-1, 1)$$

Cotangent function



$$f(x) = \cot(x)$$

$$\text{Dom } (f) = \mathbb{R} - \{n\pi, n \in \mathbb{Z}\}$$

$$\text{Ran } (f) = \mathbb{R}$$